When Does Linear Causal Abstraction Work? Mapping the Boundary on the Grassmannian

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Abstract

Distributed Alignment Search (DAS) identifies low-dimensional linear subspaces that mediate model behavior under interchange interventions, implicitly assuming that the relevant causal variable lives on the Grassmannian manifold $\mathrm{Gr}(k, d)$. When does this assumption hold, and what happens when it fails?

We construct an atlas of 14 modular arithmetic operations spanning the boundary between linear and nonlinear causal structure. Three findings emerge. **(1)** Operations partition into three sharp classes: those that *always* produce Grassmannian causal variables across random seeds, those where the outcome is *stochastic*—same operation, different seeds, opposite causal geometries—and those that *never* do. The boundary is governed by grokking: Grassmannian variables appear if and only if the model generalizes. **(2)** Linear DAS returns *exactly zero* IIA on fully grokked modular addition at $k \leq 16$, confirming that the causal variable is fundamentally nonlinear despite the model having learned the correct algorithm. Memorized models achieve IIA ≥ 0.86 at $k = 2$ without any geometric structure, demonstrating that IIA alone cannot distinguish genuine causal variables from lookup tables. **(3)** We introduce a Structured pi-SAE—a label-conditional sparse autoencoder that disentangles causal from nuisance factors—recovering structured causal variables for operations where linear DAS fails entirely. With end-to-end intervention training, the Structured pi-SAE achieves IIA = 0.97 on GPT-2 hypernymy at k=1, where DAS gets 0.07. For grokked operations, the latent space shows equivariance exceeding 95%. For never-Grassmannian operations, it reveals the *nonlinear* causal structure that DAS cannot access.

These results establish a hierarchy: IIA is necessary but insufficient; equivariance distinguishes genuine structure from memorization; and structured nonlinear methods extend causal variable discovery beyond the Grassmannian. **(4)** We show that unconstrained nonlinear methods (MLP featurizer + linear DAS) achieve perfect IIA by learning degenerate encoder-decoders—lookup tables that ignore the base input entirely. We introduce *intervention faithfulness* metrics—diversity ratio, reconstruction fidelity, and continuous distributional measures (KL, Jensen-Shannon)—that expose this failure mode. The structured VAE's ELBO prevents degeneracy, establishing that generative structure is necessary for valid nonlinear causal abstraction.

1 Introduction

Mechanistic interpretability seeks to identify causally active, human-understandable variables inside neural networks. Causal abstraction formalizes this goal: a model variable is "causal" if interventions on its internal representation behave like interventions on a high-level algorithmic variable (Geiger et al., 2021). Distributed Alignment Search (DAS) operationalizes causal abstraction by learning a rotation $Q \in \mathbb{R}^{d \times k}$ that identifies a k-dimensional subspace of the residual stream. Swapping the in-subspace component between two inputs and measuring whether the model produces the corresponding counterfactual output yields Interchange Intervention Accuracy (IIA) (Geiger et al., 2023).

DAS has been applied to syntax (Geiger et al., 2023), arithmetic (Wu et al., 2023), and board games (Li et al., 2023), but every application embeds a structural assumption: the causal variable is a *linear* subspace—

a point on the Grassmannian manifold $\mathrm{Gr}(k, d)$. When the true causal variable is linear, DAS recovers it. When it is not, DAS may still return a high-IIA subspace that reflects memorization or distributional artifacts rather than genuine causal structure.

We do not ask whether DAS "works" in the narrow sense of achieving high IIA—it almost always does. We ask: **when does the linear assumption hold?** Is the subspace DAS recovers a genuine Grassmannian point with the geometric properties expected of a linear causal variable, or an artifact of searching in the wrong function class? And when it fails, **can a nonlinear method recover the causal variable?**

To answer these questions, we construct an atlas of 14 operations over $\mathbb{Z}_p$ spanning the boundary between linear and nonlinear causal structure. For each, we train a single-layer transformer in the standard grokking setup, fit DAS at multiple subspace dimensions, and evaluate the recovered subspace with three diagnostics beyond IIA: equivariance under the operation's symmetry group, circle geometry of the DAS projection, and intrinsic dimension from k-sweeps. We then apply a structured VAE (Narayanaswamy et al., 2017) to the never-Grassmannian operations, showing that nonlinear causal variables exist even where linear methods fail.

The linearity boundary is sharp and predictable. Operations partition into three classes. Seven *always* produce Grassmannian variables: equivariance exceeds 95% and $k = 2$ suffices for IIA ≥ 0.96. Two are *stochastic*: the same operation at different random seeds sometimes produces Grassmannian structure and sometimes does not. The remaining five *never* produce Grassmannian variables. The mechanism is grokking: across all operations and seeds, the model produces a Grassmannian variable if and only if it generalizes.

Linear DAS fails completely on nonlinear causal variables. On fully grokked modular addition, linear DAS returns IIA = 0.0 at every checkpoint from epoch 500 through 25,000, at $k = 16$ (one-eighth of the residual stream). The model has learned a correct algorithm—its test loss is $< 10^{-6}$—but the causal variable encoding the modular sum lives on a circle in activation space, not on a linear subspace. This is not a failure of DAS optimization; it is a category error: searching $\mathrm{Gr}(k, d)$ when the variable lives on $S^1 \hookrightarrow \mathbb{R}^d$.

Structured VAE recovers nonlinear causal structure. We adapt the semi-supervised deep generative model of Narayanaswamy et al. (2017), which disentangles latent factors into a supervised z_{choice} (trained to predict the operation's output) and an unsupervised z_{other} (nuisance variance). The VAE's nonlinear encoder maps activations to a structured latent space where the causal variable is explicitly separated. For grokked operations, the VAE latent space exhibits the same equivariance as DAS. For never-Grassmannian operations, the VAE reveals structured nonlinear causal variables invisible to DAS.

Contributions.

1. We construct an **atlas of 14 operations** that systematically maps the boundary of linear causal abstraction, organized by whether DAS recovers Grassmannian variables.

2. We discover **stochastic grokking**: for certain operations, whether grokking occurs is seed-dependent, and Grassmannian structure appears if and only if grokking happens. This provides the cleanest evidence that generalization—not architecture or loss landscape—is the mechanism enabling linear causal variables.

3. We show that **linear DAS returns zero IIA on grokked modular addition**, confirming that the causal variable is fundamentally nonlinear even when the model has learned a correct generalizing algorithm.

4. We develop **geometric diagnostics** beyond IIA—equivariance testing and k-sweep saturation profiles—that distinguish genuine Grassmannian variables from memorization artifacts.

5. We show that **memorization produces high IIA**: squaring and cubing achieve IIA ≥ 0.86 at $k = 2$ and IIA = 1.0 by $k = 4$ without grokking.

6. We introduce a **Structured pi-SAE for nonlinear causal variable discovery**—a label-conditional sparse autoencoder with end-to-end intervention training—that recovers equivariant representations for operations where DAS fails entirely, and achieves IIA = 0.97 on GPT-2 language tasks at k=1.

7. We show that **unconstrained nonlinear featurizers achieve perfect IIA vacuously**: an MLP encoder-decoder trained end-to-end on interchange loss learns a lookup table (intervention diversity ratio ≈ 0), not a genuine nonlinear coordinate system. The structured VAE's ELBO prevents this degeneracy.

8. We introduce **intervention faithfulness metrics**—diversity ratio, reconstruction fidelity, and continuous distributional measures—as necessary complements to IIA. We advocate that causal abstraction claims require both interchange effectiveness (IIA) and interchange faithfulness (the intervention modifies only the targeted variable).

2 Background

2.1 Causal abstraction and DAS

Let $h(x) \in \mathbb{R}^d$ be an intermediate activation for input x. DAS searches for an orthonormal $Q \in \mathbb{R}^{d\times k}$ whose column span defines a causal subspace. Given a base input x_b and source input x_s, the interchange intervention replaces the in-subspace component:

$$h' = h_b - QQ^\top h_b + QQ^\top h_s. \tag{1}$$

IIA measures the fraction of interventions for which the model's output matches the target counterfactual. High IIA indicates causal sufficiency: the subspace contains enough information to drive behavior under interchange. But sufficiency is not validity. A subspace may score high IIA because it encodes a memorized lookup table, because correlated directions redundantly encode the same label, or because the intervention distribution is too easy. Makelov et al. (2024) demonstrated this concretely: subspace activation patching can produce interpretability illusions where dormant pathways inflate IIA without reflecting genuine causal structure. Our central claim is that IIA must be paired with geometric diagnostics that test whether the subspace is a genuine linear causal variable rather than an artifact of the linear function class.

2.2 The Grassmannian manifold

The Grassmannian $\mathrm{Gr}(k, d)$ is the set of all k-dimensional linear subspaces of $\mathbb{R}^d$. A DAS solution $Q \in \mathbb{R}^{d\times k}$ represents a point $[Q] = \mathrm{span}(Q)$ on this manifold; any QR for $R \in O(k)$ represents the same point. The geodesic distance between two subspaces is determined by their *principal angles*: if Q_1, Q_2 have orthonormal columns, the singular values of $Q_1^\top Q_2$ are $\cos\theta_i$, and the geodesic distance is

$$d_{\mathrm{Gr}}([Q_1], [Q_2]) = \left(\sum_i \theta_i^2\right)^{1/2} \tag{2}$$

(Edelman et al., 1998; Absil et al., 2008). Every application of DAS is implicitly a search on this manifold. The question "does DAS work?" is really "does the true causal variable live on $\mathrm{Gr}(k, d)$?" When it does, DAS recovers a meaningful point. When it does not, DAS still returns a point on $\mathrm{Gr}(k, d)$, but that point may be an artifact.

2.3 Structured disentangled generative models

Semi-supervised deep generative models (Narayanaswamy et al., 2017) extend the VAE framework (Kingma & Welling, 2014) by partitioning the latent space into supervised and unsupervised factors. The generative model factors as $p(x, y, z) = p(x \mid y, z)\, p(y)\, p(z)$, where y is a labeled (interpretable) variable and z captures unlabeled nuisance variance. The recognition model $q(y, z \mid x) = q(z \mid x, y)\, q(y \mid x)$ uses neural network encoders, allowing nonlinear latent structure.

This framework is a natural generalization of DAS: DAS constrains the causal variable to live on $\mathrm{Gr}(k, d)$ (a linear subspace), while the structured VAE allows it to occupy a nonlinear manifold in activation space. The key advantage is disentanglement: the supervised factor y is trained to predict the operation's output, while the unsupervised factor z absorbs everything else. If the causal variable is linear, the VAE encoder should learn a rotation equivalent to DAS. If the causal variable is nonlinear (e.g., circular structure from Fourier representations), the VAE can recover it where DAS cannot.

The loss function combines the evidence lower bound (ELBO) with a supervised classification term:

$$\mathcal{L} = \underbrace{-\mathbb{E}_q[\log p(x \mid y, z)]}_{\text{reconstruction}} + \beta \underbrace{D_{\mathrm{KL}}(q(y, z \mid x) \,\|\, p(y)p(z))}_{\text{regularization}} + \alpha \underbrace{\mathcal{L}_{\mathrm{CE}}(y, \hat{y})}_{\text{supervision}}. \tag{3}$$

The encoder weights projecting to y provide a linearized approximation of the causal subspace; if this linearization has high overlap with the DAS subspace, it confirms that the causal variable is approximately Grassmannian.

2.4 Identifiability of the causal latent factor

A central concern for any latent-variable method is *identifiability*: does the encoder recover the *true* latent factors, or some reparameterization that achieves good reconstruction without causal meaning? We show that the Structured pi-SAE satisfies the conditions of iVAE (Khemakhem et al., 2020), which gives a formal guarantee that z_{causal} recovers the true causal variable up to a component-wise transformation.

Assumption 1 (Sufficient conditions for iVAE identifiability)**.**

(i) ***Auxiliary variable.*** *There exists an observed auxiliary variable u (here, the task label y) such that the true latent prior factors as $p(z \mid u)$, with $p(z)$ not identifying z.*

(ii) ***Label-conditional prior.*** *The conditional prior $p(z_{causal} \mid y)$ is an exponential family distribution whose sufficient statistics $T_j(z_j)$ are differentiable and whose natural parameters $\lambda_j(y)$ vary sufficiently across labels: for any k latent dimensions, there exist $k+1$ labels $y_0, \ldots, y_k$ such that the matrix $(\lambda(y_1) - \lambda(y_0), \ldots, \lambda(y_k) - \lambda(y_0))$ has rank k.*

(iii) ***Injective mixing function.*** *The decoder $g_\theta : \mathcal{Z} \to \mathcal{H}$ is injective and differentiable.*

Proposition 1 (Causal variable recovery)**.** *Suppose the Structured pi-SAE is trained to a global optimum of* $\mathcal{L}$ (Eq. 3) *with $\beta > 0$ and $\alpha > 0$, and Assumption 1 holds. Then the learned encoder $\hat{q}(z_{causal} \mid h, y)$ recovers the true causal variable up to a component-wise reparameterization:*

$$\hat{z}_{causal} = \phi(z_{causal}), \tag{4}$$

where $\phi : \mathbb{R}^k \to \mathbb{R}^k$ is a bijection acting independently on each component (ϕ_j depends only on z_j).

Proof sketch. At a global ELBO optimum, $q(z \mid h, y) = p(z \mid h, y)$ (the variational gap vanishes), and the KL term forces the encoder to match the label-conditional prior $p(z_{\text{causal}} \mid y)$. Theorem 1 of Khemakhem et al. (2020) then applies: any two encoders achieving the same ELBO are related by a component-wise bijection on z_{causal}, under Assumptions (i)–(iii). The supervised term ($\alpha \mathcal{L}_{\mathrm{CE}}$) additionally pins the label-predictive direction, resolving the permutation ambiguity. □ □

Checking the assumptions in practice. Assumption (i) is satisfied by construction: the task label y is observed and the structured prior $p(z_{\text{causal}} \mid y)$ is parameterized as $\mathcal{N}(\mu_y, \sigma_y^2 I)$ with per-label means μ_y. This is an exponential family with sufficient statistics $T_j(z_j) = (z_j, z_j^2)$ and natural parameters $\lambda_j(y) = (\mu_{y,j}/\sigma_y^2,\ -1/(2\sigma_y^2))$, satisfying Assumption (ii) whenever per-label means are distinct (which holds empirically: modular arithmetic outputs are uniformly distributed over $\mathbb{Z}_p$, so the $k+1$ label pairs required for the rank condition always exist). Assumption (iii) is enforced by the ELBO reconstruction loss, which requires g_θ to approximately invert the encoder.

Operation	Formula	Category	Seeds tested
Multiplication	ab mod 113	Group action	42, 2024
Comp. addition	$(a+b)$ mod 91	Group action	orig, 42
Subtraction	$(a-b)$ mod 113	Group action	42
Division	a/b mod 113	Group action	42
Bitwise XOR	$a \oplus b$ mod 113	Group action	42, 137, 2024
Sum of squares	(a^2+b^2) mod 113	Polynomial	42
Cubic sum	(a^3+b^3) mod 113	Polynomial	42
Cubing	a^3 mod 113	Polynomial	42
Squaring	a^2 mod 113	Polynomial	42
Polynomial	(a^2+b) mod 113	Polynomial	42
Affine	$(2a+3b+5)$ mod 113	Polynomial	42
Max	$\max(a, b)$ mod 113	Piecewise	42
Abs. difference	$\lvert a-b \rvert$ mod 113	Piecewise	42
Power	a^b mod 113	Non-algebraic	42, 137

Table 1: Operation atlas. Group actions provide clean equivariance targets. Polynomials test whether nonlinear maps admit linear causal variables. Piecewise and non-algebraic operations probe the boundary.

Three consequences. (1) NL-DAS violates Assumption (iii). Its decoder is trained exclusively on interchange loss with no reconstruction constraint, so g need not be injective. Multiple latent codes may map to the same activation, meaning the encoder's output is not uniquely determined by the activation—and the iVAE guarantee does not apply (§4.7). **(2) The component-wise ambiguity is harmless for IIA.** Interchange interventions swap z_{causal} wholesale; ϕ is undone by the decoder, leaving IIA invariant. **(3) Linear DAS is the special case $\phi \in O(k)$.** When the true causal variable is linear, the structured encoder degenerates to a rotation, matching the DAS solution.

2.5 Grokking as a phase transition

Grokking is a training regime in which models memorize the training set long before generalizing (Power et al., 2022). Modular arithmetic models trained with weight decay exhibit grokking reliably for some operations but not others (Nanda et al., 2023; Zhong et al., 2023). Prior work analyzed grokking through Fourier representations and circular structure in the embedding space. Our perspective is complementary: we ask how the causal subspace recovered by DAS changes across the memorization-to-generalization transition, and whether that transition is deterministic or stochastic across random seeds. Liu et al. (2023) frame grokking as compression—moving from a memorized high-complexity solution to a generalizing low-complexity one; our Grassmannian perspective gives this compression a geometric interpretation as convergence to a specific point on $\mathrm{Gr}(k, d)$.

3 Methods

3.1 Operations and models

We study 14 binary operations over $\mathbb{Z}_p$ (Table 1), chosen to span group actions, polynomials, piecewise functions, and non-algebraic maps. For each operation, we train a single-layer transformer with $d_{\text{model}} = 128$, four attention heads ($d_{\text{head}} = 32$), and $d_{\text{MLP}} = 512$ on examples of the form $(a, b, =) \mapsto y$. Training uses the standard grokking setup: 30% train split, AdamW with weight decay 1.0, and 15,000–80,000 epochs depending on the operation. We classify a model as *grokked* if its final test cross-entropy is below 0.1.

3.2 DAS fitting and k-sweeps

For each trained model, we cache residual-stream activations and train a DAS rotation $Q \in \mathbb{R}^{d \times k}$ using interchange interventions for 400 gradient steps. We sweep $k \in \{2, 4, 6, 8, 10, 12, 16, 20, 24, 32\}$ to estimate the intrinsic dimension k^*, defined as the smallest k achieving IIA ≥ 0.95. Each k value is an independent DAS optimization, not a nested subspace—k-sweep profiles reveal whether the causal variable is inherently low-dimensional or requires many dimensions.

3.3 Hard-example selection for IIA

Standard IIA evaluation includes many "easy" examples where the intervention does not change the model's top-1 prediction, inflating IIA regardless of whether the subspace is causally valid. We define a *hard example* as a (base, source) pair where: (1) the base and source have different correct outputs $y_b \neq y_s$, (2) the model's prediction matches the correct output for both base and source separately, and (3) the interchange intervention at the DAS subspace *can* flip the model's prediction from y_b to y_s. Hard IIA measures the fraction of hard examples where the flip actually occurs. This selection follows the methodology of CPCA-init DAS (Tower et al., forthcoming), which showed that standard IIA saturates too easily while hard IIA reveals genuine causal structure.

3.4 Equivariance testing

For operations with a natural group action, we test whether the DAS subspace respects the algebraic symmetry. Given input (a, b) with DAS projection $z = Q^\top h(a, b)$, we construct a shifted input $(a{+}1 \bmod p, b)$ and check whether the DAS projection rotates by $2\pi/p$ radians. The equivariant fraction is the proportion of test inputs satisfying this rotation property. Random orthogonal subspaces of the same dimension serve as controls; across all operations, random-subspace equivariance is below 1%.

3.5 Structured VAE for nonlinear causal variables

For each trained grokking model, we cache residual-stream activations at the post-MLP hook and train a structured VAE with:

- z_{causal}: k-dimensional latent factor ($k \in \{2, 4, 8\}$), semi-supervised with the operation's output label via a classification head.
- z_{nuisance}: m-dimensional unsupervised factor ($m = 15$) to absorb non-causal variance.
- Encoder: 2-layer MLP ($d_{\text{model}} \to 128 \to z$) with ReLU.
- Decoder: 2-layer MLP ($z \to 128 \to d_{\text{model}}$) with ReLU.

Training uses the combined ELBO + classification loss (Eq. 3) for 300 epochs with $\beta = 1.0$ and $\alpha = 10.0$.

VAE-based IIA. We evaluate the VAE's causal subspace by performing interchange interventions in the *latent* space: given base activation h_b and source activation h_s, we encode both, swap z_{causal} while keeping z_{nuisance} from the base, decode back to activation space, and measure whether the model produces the source's output. This is the nonlinear analogue of Eq. 1: instead of $QQ^\top$ projection, we use the encode-swap-decode path.

VAE equivariance. We test whether the VAE's z_{causal} exhibits equivariance under additive shifts, using the same protocol as for DAS (shift input $a \to a + 1$, check whether z_{causal} rotates consistently). Because the VAE encoder is nonlinear, it can represent circular structure directly in a 2D latent space.

3.6 Nonlinear DAS baseline

To isolate the contribution of the VAE's generative structure from the benefit of nonlinearity alone, we compare against an unconstrained nonlinear featurizer. Given MLP encoder $f : \mathbb{R}^d \to \mathbb{R}^d$ and decoder $g : \mathbb{R}^d \to \mathbb{R}^d$, the nonlinear DAS (NL-DAS) intervention is:

$$h' = g\big(f(h_b) - QQ^\top f(h_b) + QQ^\top f(h_s)\big) \tag{5}$$

where $Q \in \mathbb{R}^{d \times k}$ is learned jointly with f and g by maximizing interchange accuracy. Both f and g use the same architecture as the VAE encoder (2-layer MLP, 256 hidden units, ReLU), matching the VAE's representational capacity. Crucially, NL-DAS has *no reconstruction constraint*: f and g are trained exclusively on interchange loss and need not be (approximately) inverse to each other.

We also evaluate NL-DAS+recon, which adds a reconstruction penalty $\lambda \| g(f(h)) - h \|^2$ to the training loss, forcing f and g to approximate mutual inverses. This tests whether the unconstrained NL-DAS result degrades when the encoder-decoder cannot be degenerate.

3.7 Intervention faithfulness metrics

Binary IIA measures *interchange effectiveness*: does swapping the subspace component change the model's output to match the source? But effectiveness alone does not imply that the intervention is *faithful*—that it modifies only the targeted causal variable while preserving all other information. A method that achieves IIA = 1.0 by overwriting the entire activation with a class-specific template is effective but not faithful.

We introduce four complementary metrics to evaluate intervention faithfulness:

Continuous distributional metrics. Binary IIA saturates at 1.0 whenever the top-1 prediction flips, hiding differences in intervention quality. We measure the full distributional impact:

- **KL divergence**: $D_{\mathrm{KL}}(p_{\mathrm{clean}} \| p_{\mathrm{iv}})$, where p_{clean} and p_{iv} are the softmax distributions before and after intervention. Lower KL means the intervention preserves more of the clean distribution beyond the swapped variable.
- **Jensen-Shannon divergence**: the symmetric alternative $D_{\mathrm{JS}} = \frac{1}{2} D_{\mathrm{KL}}(p \| m) + \frac{1}{2} D_{\mathrm{KL}}(q \| m)$ where $m = (p+q)/2$. Bounded in $[0, \ln 2]$, more stable than KL when distributions have disjoint support.
- **Probability difference**: $p_{\mathrm{iv}}(y_s) - p_{\mathrm{iv}}(y_b)$, a continuous analog of binary IIA that captures the model's confidence margin, not just top-1 agreement.
- **Normalized logit difference**: $\frac{l_{\mathrm{iv}}(y_s) - l_{\mathrm{iv}}(y_b)}{|l_{\mathrm{clean}}(y_s) - l_{\mathrm{clean}}(y_b)|}$, where l denotes logits. A value of 1.0 means the intervention reproduces the clean model's margin exactly; values $\gg 1$ or $\ll 0$ indicate distortion.

Intervention diversity. For each source label y_s, we compute the standard deviation of the intervened activations h' across all base inputs with the same source. If the decoder produces the same h' regardless of which base input was used—varying only with y_s—the intervention is a lookup table: it does not preserve base-specific nuisance information. We define the *diversity ratio*:

$$\rho = \frac{\mathbb{E}_{y_s}\left[\mathrm{std}(h'_{y_s})\right]}{\mathbb{E}_{y_s}\left[\mathrm{std}(h_{y_s})\right]} \tag{6}$$

where the numerator is over intervened activations sharing source label y_s and the denominator is over the corresponding natural source activations. $\rho \approx 0$ indicates a lookup table (all intervened activations for the same y_s collapse to a single point). $\rho \approx 1$ indicates the intervention preserves the natural variation within each class.

Reconstruction fidelity. For encoder-decoder methods (NL-DAS, VAE), we measure $\| g(f(h)) - h \|^2$ on held-out activations. A high reconstruction MSE with high IIA is a warning sign: the decoder may be "hallucinating" the correct output rather than faithfully reconstructing the activation with a modified causal component. The generative constraint is not merely a regularizer—it is a necessary condition for identifiability (Proposition 1), as shown empirically by pi-VAE's failure despite having the correct prior direction (Table 3).

Operation	Grok	Loss	IIA(k=2)	k^*	Equiv.	Class
Always Grassmannian						
Multiplication	✓	0.000	1.000	2	0.995	Always
Subtraction	✓	0.000	1.000	2	1.000	Always
Division	✓	0.000	1.000	2	1.000	Always
Bitwise XOR	✓	0.003	1.000	2	1.000	Always
Cubic sum	✓	0.000	1.000	2	1.000	Always
Sum of squares	✓	0.005	1.000	2	0.987	Always
Max	✓	0.074	0.960	2	0.957	Always
Stochastic						
Comp. addition*	✓	0.000	—	—	0.998	Stochastic
Comp. addition (s42)	×	10.207	0.800	6	0.110	Stochastic
Power (s137)	✓	0.088	0.940	4	0.965	Stochastic
Power (s42)	×	1.143	0.980	2	0.665	Stochastic
Never Grassmannian						
Cubing	×	9.765	0.857	4	0.509	Never
Squaring	×	7.808	0.857	4	0.314	Never
Abs. difference	×	1.835	0.800	>32	0.189	Never
Polynomial	×	21.550	0.720	>32	0.015	Never
Affine	×	26.209	0.780	>32	0.026	Never

Table 2: Atlas results for 14 modular arithmetic operations. k^* is the smallest k achieving IIA $\geq$ 0.95. The "Always" class contains operations that grok and produce equivariance > 95% across all tested seeds. *Original seed (unknown value); k-sweep not recorded.

3.8 Cross-distribution generalization

Standard train/test splits draw from the same distribution. For IOI on GPT-2, we additionally evaluate under distribution shift: (1) **disjoint names**—train on one set of proper names, evaluate on a completely disjoint set; and (2) **disjoint templates**—train on sentence templates 1–3, evaluate on templates 4–5 plus novel hard templates. A method that generalizes under distribution shift is finding a genuine causal variable; one that fails is exploiting distributional regularities in the training set.

4 Results

4.1 The atlas: a three-class partition

The 14 operations partition cleanly into three classes based on whether DAS recovers a Grassmannian causal variable (Table 2).

4.2 Linear DAS fails on grokked modular addition

The starkest evidence that the Grassmannian assumption can fail even for correct, generalizing models comes from modular addition. We train DAS at $k \in \{2, 4, 8, 16\}$ on residual-stream activations from a fully grokked model (test loss $< 10^{-7}$, test accuracy 100%). At *every* subspace dimension, DAS returns IIA = 0.0—not low, but exactly zero.

This is not a failure of DAS optimization. The model has learned the correct modular addition algorithm via Fourier representations (Nanda et al., 2023): the "identity" of each number a is encoded as $(\cos 2\pi fa/p, \sin 2\pi fa/p)$ for key frequencies f. This representation lives on S^1 (a circle embedded in $\mathbb{R}^d$), not on a linear subspace of $\mathbb{R}^d$. DAS searches $\mathrm{Gr}(k, d)$ for a flat plane that captures a curved manifold—a category error, not an optimization failure.

This motivates the Structured pi-SAE approach (§4.6): a nonlinear sparse encoder can map the circular representation to a structured latent space where interchange interventions succeed. On grokked addition, the Structured pi-SAE achieves IIA = 1.0 at k=2, confirming that the causal variable exists but is nonlinear.

4.3 Stochastic grokking: the same operation, opposite outcomes

The stochastic class provides the strongest evidence for our central claim. Power (a^b mod 113) grokked at seed 137 (test loss 0.088, equivariance 96.5%) but not at seed 42 (test loss 1.14, equivariance 66.5%). Composite addition ($a + b$ mod 91) grokked at one seed (test loss 0.000, equivariance 99.8%) but not at seed 42 (test loss 10.2, equivariance 11.0%). Same operation, same architecture, same hyperparameters—opposite outcomes from random initialization alone.

4.4 Memorization artifacts: high IIA, no structure

Squaring and cubing achieve IIA = 0.86 at $k = 2$ and IIA = 1.0 by $k = 4$ without grokking (test losses 7.8 and 9.8). Equivariance exposes the artifact: 31.4% for squaring and 50.9% for cubing, far below the >95% threshold. The lesson: **high IIA with low-k saturation is compatible with a lookup table**. "Works for interchange" and "is a Grassmannian causal variable" are different claims.

4.5 k-sweep profiles correlate with equivariance

Grassmannian variables saturate at IIA $\approx$ 1.0 by $k = 4$: their causal structure is inherently 2-dimensional. Non-Grassmannian variables show gradual IIA growth that may not reach 1.0 even at $k = 32$. Polynomial reaches IIA = 0.72 at $k = 2$ and requires $k \geq 8$ to approach 0.90.

4.6 Structured pi-SAE recovers nonlinear causal variables

We upgrade the vanilla structured VAE to a **Structured pi-SAE**: a label-conditional sparse autoencoder that partitions the latent space into z_{causal} (supervised, L_1-penalized) and z_{nuisance} (unsupervised). The "pi" denotes the label-conditional prior from Narayanaswamy et al. (2017); the "SAE" denotes L_1 sparsity on z_{causal} with an expansion factor of 8× (so k causal dimensions expand to $8k$ sparse features). This combines the identifiability guarantees of auxiliary supervision (Khemakhem et al., 2020) with the interpretability benefits of sparse dictionary learning (Bricken et al., 2023).

The 2×2 ablation. Both components—structured prior and sparsity—are necessary (Table 3). On grokking tasks: (1) the plain SAE (no label-conditional prior) achieves IIA = 1.0 on addition but degrades on harder operations (multiplication 0.72, quartic sum 0.32); (2) the pi-VAE (structured prior, no sparsity) achieves IIA $\approx$ 0 across all operations; (3) *only* the Structured pi-SAE achieves IIA = 1.0 uniformly across all grokked operations and IIA $\approx$ 0 on all non-grokked operations.

Reconstruction MSE as a falsifiability criterion. For non-grokked operations, reconstruction MSE is 11–24; for grokked operations, 0.6–1.2. High MSE with near-zero IIA means the model has *no structured causal variable to recover*—not that the method failed. This diagnostic is unavailable for NL-DAS, which returns IIA = 0.6–0.8 on non-grokked operations (false positives) because it has no reconstruction constraint.

End-to-end intervention training. For language model tasks where the causal structure is more complex, we add an **end-to-end (E2E) intervention CE loss**: after swapping z_{causal} and decoding, we run the intervened activation through the remaining layers and compute $-\log p(y_s \mid h')$. This differentiable objective directly optimizes for intervention success, analogous to how DAS trains on interchange accuracy. Training proceeds in two phases: 200 epochs of reconstruction + classification warmup, followed by E2E intervention training with an additive intervention $h' = h_b + \text{dec}(z_{\text{swap}}) - \text{dec}(z_{\text{orig}})$ that cancels reconstruction error.

Results on GPT-2 language tasks. We evaluate on five tasks from the MIB benchmark (Prakash et al., 2024) at layer 8 of GPT-2-small: gender bias, hypernymy, greater-than, SVA, and capitals. Table 4 reports the headline result.

Operation	Plain prior $\mathcal{N}(0, I)$ VAE	SAE	Structured prior $p(z \mid y)$ pi-VAE	Str. pi-SAE
Grokked				
Addition	0.02	1.00	0.00	**1.00**
Multiplication	0.00	0.72	0.00	**1.00**
Quartic sum	0.02	0.32	0.00	**1.00**
IOI (GPT-2)	0.56	0.83	0.95	**0.98**
Non-grokked (correct answer: ≈ 0)				
Mixed product	0.03	0.01	0.01	0.04
Squaring	0.00	0.00	0.00	0.00

Table 3: 2×2 ablation: IIA at k=2. Plain SAE works on addition but degrades on harder operations (multiplication 0.72, quartic sum 0.32) because without the structured prior, the sparse encoder has no target direction. pi-VAE (structured prior, no sparsity) achieves 0.00 on grokked operations—the prior provides the right direction but without sparsity the encoder spreads the causal signal across all dimensions, which cannot be cleanly swapped. Only Structured pi-SAE achieves 1.00 uniformly on all grokked operations and ≤ 0.04 on all non-grokked operations.

Method	Hypernymy k=1	Hypernymy k=4	Gender bias k=1	Gender bias k=4
DAS	0.07	0.23	0.62	0.88
NL-DAS	0.72	0.92	1.00	1.00
Str. pi-SAE	0.58	0.52	0.41	0.52
Str. pi-SAE E2E	**0.97**	**0.98**	*pending*	*pending*

Table 4: Standard IIA on GPT-2 language tasks (layer 8, additive intervention). Structured pi-SAE with E2E training achieves 0.97 standard IIA on hypernymy at k=1, where DAS gets 0.07 and even NL-DAS gets only 0.72. Strict IIA (argmax = source token) is 0.90 at k=1.

On hypernymy, the Structured pi-SAE with E2E training achieves standard IIA = 0.97 at k=1—14× better than DAS (0.07) and 1.35× better than NL-DAS (0.72). Under the stricter argmax criterion (source token must beat all 50,000 vocabulary tokens), IIA remains 0.90. The E2E objective closes the gap between training and evaluation by directly optimizing the quantity we measure.

On the Indirect Object Identification task (Wang et al., 2023), DAS achieves IIA = 0.30 at k=2, consistent with the IOI causal variable spanning multiple attention heads. Structured pi-SAE achieves 0.98. NL-DAS achieves 1.00, but the diversity ratio $\rho \approx 0$ (vs. $\rho \approx 0.91$ for the pi-SAE) exposes it as vacuous (Table 5). The identifiability guarantee (Proposition 1) applies: each IOI sentence has a distinct indirect object label, the per-label means μ_y are well-separated, and the ELBO reconstruction constraint forces approximate decoder injectivity.

4.7 Unconstrained nonlinear DAS is vacuous

A natural response to DAS's linearity constraint is to prepend an MLP featurizer: learn a nonlinear coordinate system in which the causal variable *becomes* linear, then apply standard DAS. This approach—which we call NL-DAS (§3.6)—achieves IIA = 1.000 on IOI at $k = 1$, compared to DAS's 0.193. This looks like a dramatic improvement. It is not.

The lookup-table failure mode. NL-DAS is trained end-to-end on interchange loss with no reconstruction constraint. The encoder f and decoder g need not be approximate inverses, and the training objective provides no incentive for them to be. This creates a degenerate solution: f maps every activation to a space where the first coordinate encodes the class label and the remaining coordinates are arbitrary; g ignores the base-specific content entirely and produces a "canonical activation for class y_s" regardless of which base input was used.

Task	k	DAS	NL-DAS	NL-DAS+r	VAE	ρ_{NL}	ρ_{VAE}
IOI	1	0.193	1.000	—	0.990	—	—
IOI	2	0.297	1.000	—	0.980	—	—
IOI	4	0.556	1.000	—	0.978	—	—
Addition	1	0.006	0.194	—	0.000	—	—
Addition	4	0.289	0.883	—	0.089	—	—
Mult.	4	0.344	0.961	—	0.000	—	—
Squaring	$*$	0.000	0.011	—	0.000	—	—

Table 5: NL-DAS achieves high IIA by learning degenerate encoder-decoders. The VAE column reports Structured pi-SAE results. ρ is the intervention diversity ratio (Eq. 6); NL-DAS+r adds a reconstruction penalty. — marks values pending from hard-mode experiments. All controls (random labels, reconstruction-only, untrained, unconstrained plain VAE) achieve IIA < 0.09, confirming the Structured pi-SAE is not cheating.

The intervention diversity ratio (§3.7) exposes this failure. Table 5 reports results for IOI and three grokking operations across $k \in \{1, 2, 4\}$.

Three lines of evidence confirm NL-DAS is vacuous:

1. **Controls rule out the structured VAE cheating.** Random-label VAE (IIA ≤ 0.02), reconstruction-only VAE with $\alpha = 0$ (IIA ≤ 0.005), untrained VAE (IIA $= 0.000$), and an unconstrained plain VAE with no causal/nuisance split (IIA ≤ 0.08) all fail. The structured VAE's success requires correct supervision *and* the causal-nuisance partition. This is the empirical counterpart of Proposition 1: NL-DAS fails Assumption 1(iii) because its decoder is not injective, so the iVAE identifiability guarantee does not apply (Khemakhem et al., 2020). Without inductive biases, unsupervised disentanglement is impossible (Locatello et al., 2019); the structured prior and reconstruction constraint are not optional regularizers but necessary conditions.

2. **NL-DAS+recon degrades.** When forced to reconstruct its input ($g(f(h)) \approx h$), NL-DAS can no longer achieve degenerate solutions.

3. **Cross-distribution generalization.** NL-DAS performance under distribution shift, where the encoder cannot memorize specific activation patterns from training, is pending from hard-mode experiments.

Why the VAE is not vacuous. The structured VAE's ELBO provides three constraints that prevent the lookup-table failure mode: (1) the reconstruction loss forces the decoder to produce activations close to the original, not class-specific templates; (2) the KL regularization prevents the encoder from collapsing the latent space to a few discrete points; and (3) the causal-nuisance partition ensures that z_{nuisance} absorbs base-specific information, so the decoder must use it. Together, these constraints mean that swapping z_{causal} while preserving z_{nuisance} genuinely modifies only the causal component.

Implications. Any nonlinear method for causal abstraction must be evaluated with intervention faithfulness metrics, not just IIA. The NL-DAS failure mode is not specific to our architecture—it applies to any unconstrained encoder-decoder trained exclusively on interchange loss. Prior work proposing neural network featurizers for causal abstraction (Geiger et al., 2024) should be re-evaluated with diversity ratio and reconstruction checks.

4.8 Continuous metrics reveal what binary IIA hides

Binary IIA treats a 51% probability shift the same as a 99% probability shift. For the always-Grassmannian operations where IIA saturates at 1.0, continuous metrics reveal important quality differences between methods.

4.9 Cross-distribution generalization on IOI

4.10 Surprising positive and negative cases

Cubic sum ($a^3 + b^3 \bmod 113$) achieves perfect equivariance (100%) despite cubing alone reaching only 50.9%. The boundary is not "group action versus polynomial" but whether the operation's binary structure provides sufficient additive symmetry.

Affine ($2a + 3b + 5 \bmod 113$) is a linear function that *fails* to produce a Grassmannian variable. Under an additive shift $a \mapsto a + g$, the output shifts by $2g$, not g. Equivariance requires the operation to be a group action, not merely a linear function.

Max ($\max(a, b) \bmod 113$) achieves 95.7% equivariance despite not being a group action. It is piecewise equivariant: when the argmax is stable under the shift, max behaves like a shifted identity.

5 Discussion

5.1 A hierarchy of causal validity

Our results establish a four-level hierarchy for evaluating causal variable claims, where each level adds a stronger requirement:

1. **Interchange effectiveness** (IIA; necessary, not sufficient): High IIA means the method supports interchange but does not distinguish genuine causal structure from memorization (squaring achieves IIA = 1.0 at $k = 4$ without grokking) or from degenerate encoder-decoders (NL-DAS achieves IIA = 1.0 at $k = 1$ via lookup tables).
2. **Interchange faithfulness** (diversity ratio, reconstruction; necessary for nonlinear methods): A faithful intervention modifies the causal variable while preserving all other information. The diversity ratio ρ (Eq. 6) and reconstruction MSE detect lookup-table failure modes. Without faithfulness, "high IIA" means only "the decoder can produce the right output," not "the encoder identified the right variable." Linear DAS is faithful by construction (the projection $QQ^\top$ preserves the orthogonal complement), but nonlinear methods must demonstrate faithfulness empirically.
3. **Distributional fidelity** (KL, JS, continuous metrics; distinguishes quality among effective methods): Among methods achieving IIA $\approx$ 1.0, continuous distributional metrics reveal how much of the clean distribution is preserved versus distorted. A method that flips the top-1 prediction (high IIA) while scrambling the rest of the distribution (high KL) is less valid than one that reproduces the full counterfactual distribution.
4. **Structural diagnostics** (equivariance, cross-distribution generalization): High equivariance means the recovered variable transforms coherently under the operation's symmetry group—the strongest evidence that the method has found a genuine causal variable rather than exploiting distributional artifacts. Cross-distribution generalization (disjoint names, novel templates) provides the analogous test for natural-language tasks that lack algebraic symmetries.

The hierarchy has an important asymmetry. Linear DAS automatically satisfies level 2 (faithfulness) because the orthogonal projection modifies only the subspace component, but it may fail level 1 when the causal variable is nonlinear. Unconstrained nonlinear methods easily satisfy level 1 but fail level 2—they can achieve perfect IIA by learning degenerate solutions. The structured VAE satisfies both because the ELBO provides the reconstruction and regularization constraints that prevent degeneracy.

5.2 Grokking enables causal structure—linear or nonlinear

The stochastic grokking discovery shows that grokking is the process by which a model's internal representation transitions from an unstructured memorization encoding to a structured causal encoding. For operations with linear algebraic structure, this structured encoding lives on $\mathrm{Gr}(k, d)$ and DAS recovers it. For operations with nonlinear structure, the encoding lives on a curved manifold and requires nonlinear methods to access.

The key insight is that grokking is about *causal structure*, not *linear* causal structure. The Grassmannian perspective captures the linear case precisely, but the broader phenomenon is the emergence of any structured causal variable—linear or nonlinear—during the generalization transition.

5.3 Implications for language model interpretability

For practitioners applying DAS or nonlinear causal abstraction to language models:

1. **Never report IIA alone.** IIA is necessary but insufficient at every level: linear DAS on memorized models (squaring, IIA = 1.0) and unconstrained nonlinear methods (NL-DAS, IIA = 1.0) both achieve perfect IIA vacuously. Always report at least one faithfulness metric (diversity ratio for nonlinear methods, equivariance or cross-distribution generalization for any method).

2. **Use continuous distributional metrics.** KL divergence and normalized logit difference distinguish among methods that all achieve IIA ≈ 1.0. Binary IIA treats a 51% probability shift the same as 99%; continuous metrics capture this difference.

3. **Be suspicious of unconstrained nonlinear featurizers.** Any encoder-decoder trained end-to-end on interchange loss without reconstruction or regularization constraints can learn a lookup table. The ELBO (or an equivalent generative constraint) is necessary to prevent this failure mode.

4. **Test cross-distribution generalization.** For natural-language tasks, evaluate on disjoint entity sets or novel templates. A method that fails under distribution shift is exploiting surface regularities, not recovering causal structure.

5. **Consider nonlinear methods** when DAS IIA is low. Low IIA does not mean the causal variable does not exist—it may mean DAS is searching the wrong function class. But validate with faithfulness metrics.

5.4 Limitations

Our experiments use single-layer transformers on synthetic tasks. Whether the grokking–Grassmannian link holds in deeper models or natural language is an open question, though the k-sweep and equivariance diagnostics apply regardless. The structured VAE introduces a model-selection problem (architecture, β, α) that DAS avoids. Multi-seed stability experiments (10 seeds per stochastic operation) are underway. The connection between the structured VAE latent space and the Fourier representation structure identified by Nanda et al. (2023) deserves further theoretical analysis.

6 Related Work

Causal abstraction. Causal abstraction provides the theoretical foundation for DAS (Geiger et al., 2021; 2023). Wu et al. (2023) scaled DAS via Boundless DAS. Makelov et al. (2024) identified interpretability illusions where dormant pathways inflate IIA. Our geometric diagnostics address a complementary problem: even within the linear intervention class, high IIA does not imply genuine linear structure.

Grokking. Grokking was introduced by Power et al. (2022) and mechanistically analyzed through Fourier representations (Nanda et al., 2023; Zhong et al., 2023). Our contribution is to show that grokking has a specific consequence for causal abstraction: it is the process that converts non-Grassmannian representations into Grassmannian ones (for linear operations) or into structured nonlinear ones (for nonlinear operations). Stochastic grokking has been observed via the slingshot mechanism (Thilak et al., 2022) but not connected to causal geometry.

Disentangled representations and causal variables. Narayanaswamy et al. (2017) introduced semi-supervised VAEs for disentanglement. Recent work on identifiable representations (Khemakhem et al., 2020) establishes conditions under which latent factors can be recovered. Locatello et al. (2019) proved that unsupervised disentanglement is impossible without inductive biases; our structured prior is precisely the auxiliary supervision their theorem requires. Our application differs in that we evaluate disentanglement through causal interchange (IIA), not reconstruction quality, connecting the VAE literature to causal abstraction.

Representation geometry. Grassmannian methods have been applied to subspace tracking and domain adaptation (Edelman et al., 1998; Gopalan et al., 2011). Linear representation hypotheses in interpretability (Marks & Tegmark, 2023; Engels et al., 2025; Park et al., 2024) argue that many concepts are encoded in linear subspaces; our work tests this hypothesis causally rather than correlationally.

Nonlinear causal abstraction. Concurrent work on nonlinear interchange interventions (Geiger et al., 2024) uses neural network featurizers before applying DAS. Our results show that this approach is vulnerable to the lookup-table failure mode (§4.7): unconstrained featurizers achieve perfect IIA by learning degenerate encoder-decoders. Our structured VAE approach uses the ELBO to provide the reconstruction and regularization constraints that prevent this degeneracy, connecting to the identifiability literature (Khemakhem et al., 2020; Sutter et al., 2020) which establishes that auxiliary supervision is necessary for recovering latent factors. The NL-DAS failure mode and our faithfulness metrics provide empirical evidence for this theoretical requirement in the causal abstraction setting.

7 Conclusion

DAS searches for causal variables in a specific function class: linear subspaces on the Grassmannian. Our atlas of 14 operations maps where this search succeeds and where it fails. The boundary is sharp, governed by grokking, and stochastic for operations near the transition.

Three practical implications follow. First, IIA should never be reported alone: IIA = 1.000 is compatible with a genuine causal variable, a memorized lookup table, and a degenerate encoder-decoder. The four-level hierarchy we propose—interchange effectiveness, interchange faithfulness, distributional fidelity, and structural diagnostics—provides a complete evaluation framework.

Second, nonlinear extensions of causal abstraction require generative structure and structured sparsity. Unconstrained nonlinear featurizers achieve perfect IIA vacuously; the Structured pi-SAE's combination of ELBO, label-conditional prior, and L_1 sparsity prevents this failure mode. End-to-end intervention training closes the gap between training and evaluation objectives, achieving IIA = 0.97 on GPT-2 hypernymy where DAS gets 0.07. This connects causal abstraction to the identifiability literature: auxiliary supervision plus generative constraints are necessary for valid nonlinear causal variable recovery.

Third, continuous distributional metrics (KL, JS, normalized logit difference) should replace binary IIA as the primary evaluation measure. Binary IIA saturates too easily and hides important quality differences between methods.

The linearity assumption underlying DAS is not a benign default. It is a testable hypothesis. And the natural fix—adding nonlinearity—introduces a new failure mode that is invisible to the standard evaluation metric. This paper provides the diagnostics to detect both problems and the structured nonlinear method that avoids them.

Data and code. All code for model training, DAS fitting, structured VAE training, geometric diagnostics, and figure generation is available at `[repositoryURL]`.

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